Researchers measured several cardiovascular parameters from the heart and aorta in patients undergoing cardiac catheterization. The values in Table 1 were measured by inserting catheters through the vascular systems of the patients and utilizing ultrasound imaging, pressure sensors, and Doppler velocity sensors. A catheter with a Doppler velocity sensor operating at an ultrasound frequency of 10 MHz was inserted into the aorta to measure blood velocity at different phases of the cardiac cycle.

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Aorta	Blood velocity (cm/s)	20	10	45
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Left ventricle	Blood pressure (mmHg)	*	10	120
	Stroke volume (mL)	90	*	*
Right ventricle	Blood pressure (mmHg)	*	5	30
	Stroke volume (mL)	90	*	*

*not measured

The pulse pressure *PP* is defined as the difference between the systolic pressure P_S and the diastolic pressure P_D:

$$\mathrm{PP} = P_S - P_D$$

Equation 1

Poiseuille's law relates volumetric blood flow *Q* through a vessel to the change in blood pressure Δ*P*, blood viscosity *η*, and physical dimensions of the vessel:

$$Q = \frac{\Delta P \pi r^4}{8\eta L}$$

Equation 2

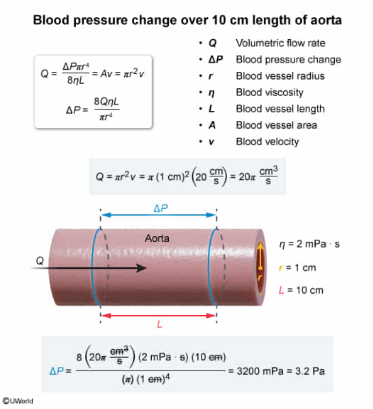
where *r* is the radius of the blood vessel and *L* is its length.

What is the change in blood pressure over a 10 cm long section of the aorta? (Note: Viscosity of blood is *η* = 2 mPa·s.)

- A. 10 Pa [14%]
- B. 3.2 Pa [46%]
- C. 1.0 Pa [24%]
- D. 0.8 Pa [15%]

Correct
47% answered correctly

Explanation:



Poiseuille's law states that the volumetric blood flow *Q* through a blood vessel depends on the change in **blood pressure** Δ*P*, vessel radius *r*, blood **viscosity** *η*, and vessel length *L* according to the following equation:

$$Q = \frac{\Delta P \pi r^4}{8\eta L}$$

This equation can be rearranged to solve for Δ*P*, yielding:

$$\Delta P = \frac{8Q\eta L}{\pi r^4}$$

In this question, *η* equals 2 mPa·s, *r* equals 1 cm (ie, half of the aortic diameter given in Table 1), and *L* equals 10 cm. *Q* represents the

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volume of blood flowing through the aorta and equals the product of the aorta's cross-sectional area *A* and the average blood **velocity** *v*:

$$Q = \mathrm{Av} = \pi r^2 v$$

According to Table 1, *v* equals 20 cm/s. Substituting the values of *r* and *v* into the above equation yields:

$$Q = \pi(1\ \mathrm{cm})^2\left(20\ \frac{\mathrm{cm}}{\mathrm{s}}\right) = 20\pi\ \frac{\mathrm{cm}^3}{\mathrm{s}}$$

Substituting the values of *Q*, *η*, *L*, and *r* into the equation for Δ*P* gives:

$$\Delta P = \frac{8\left(20\cancel{\pi}\ \frac{\cancel{\mathrm{cm}^3}}{\mathrm{s}}\right)(2\ \mathrm{mPa} \bullet \mathrm{s})(10\ \cancel{\mathrm{cm}})}{\cancel{\pi}(1\ \cancel{\mathrm{cm}})^4}$$

$$\Delta P = 3200\ \mathrm{mPa} = 3.2\ \mathrm{Pa}$$

Therefore, Δ*P* over a 10 cm long section of the aorta is 3.2 Pa.

(Choice A) 10 Pa is obtained if the *π* term is not included in the denominator of the Δ*P* equation.

(Choice C) 1 Pa results from using 20 m/s for the volumetric flow rate.

(Choice D) 0.8 Pa is calculated if 2 cm is used for the radius of the aorta instead of 1 cm.

Educational objective:
Poiseuille's law describes fluid flow through a cylindrical pipe in terms of the volumetric flow rate, change in pressure, fluid viscosity, and physical dimensions of the pipe. The equation can be used to calculate the change in blood pressure over a specific length of the aorta.

Time spent: 0 sec
Subject: Physics

QID: 403694
System: Fluids

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	Stroke volume (mL)	90	*	*
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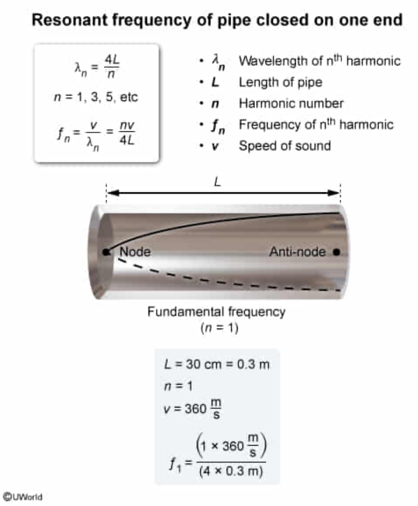
where *r* is the radius of the blood vessel and *L* is its length.

Assume the aorta is a straight, uniform pipe closed at one end. Which expression gives the fundamental resonant frequency (in Hz) for this pipe? (Note: Assume the speed of sound in air *v* = 360 m/s.)

- A. (1 × 360) / (4 × 30) [7%]
- B. (1 × 360) / (2 × 30) [10%]
- ✔ C. (1 × 360) / (4 × 0.3) [60%]
- D. (1 × 360) / (2 × 0.3) [21%]

Correct
60% answered correctly

Explanation:



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The standing waves that form in a pipe consist of specific wavelengths λ based on the fundamental harmonic for the pipe. For a pipe closed on one end, each harmonic wavelength λ_n is calculated from the ratio of the **length** L of the pipe and the **harmonic number** n using the equation:

$$\lambda_n = \frac{4L}{n}$$

where n is any odd, positive integer (eg, 1, 3, 5, etc.). The fundamental harmonic is given by λ_1 (ie, n = 1):

$$\lambda_1 = 4L$$

Furthermore, the harmonic frequency f_n equals the ratio of the speed of sound v and λ_n:

$$f_{\mathrm{n}} = \frac{v}{\lambda_n} = \frac{\mathrm{nv}}{4L}$$

Thus, the fundamental harmonic frequency, n = 1, equals:

$$f_1 = \frac{v}{4L}$$

In this question, the aorta is a pipe with L equal to 30 cm (ie, 0.3 m) and v equal to 360 m/s. Therefore, f_1 equals:

$$f_1 = \frac{(1 \times 360)}{(4 \times 0.3)}$$

(Choices A and B) The 30 term in the denominator is the length of the pipe in cm, but this value needs to be converted into m.

(Choice D) The term (2 × 0.3) in the denominator uses a factor of 2, but a factor of 4 is required because the pipe is closed on one end.

Educational objective:
The harmonic frequency in a pipe equals the ratio of the speed of sound and the harmonic wavelength. The fundamental harmonic wavelength of a pipe closed on one end is four times greater than the pipe's length.

Time spent:0 sec
Subject:
Physics

QID:403695
System:
Fluids

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Researchers measured several cardiovascular parameters from the heart and aorta in patients undergoing cardiac catheterization. The values in Table 1 were measured by inserting catheters through the vascular systems of the patients and utilizing ultrasound imaging, pressure sensors, and Doppler velocity sensors. A catheter with a Doppler velocity sensor operating at an ultrasound frequency of 10 MHz was inserted into the aorta to measure blood velocity at different phases of the cardiac cycle.

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	Diameter (cm)	2	*	*
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Left ventricle	Blood pressure (mmHg)	*	10	120
	Stroke volume (mL)	90	*	*
Right ventricle	Blood pressure (mmHg)	*	5	30
	Stroke volume (mL)	90	*	*

*not measured

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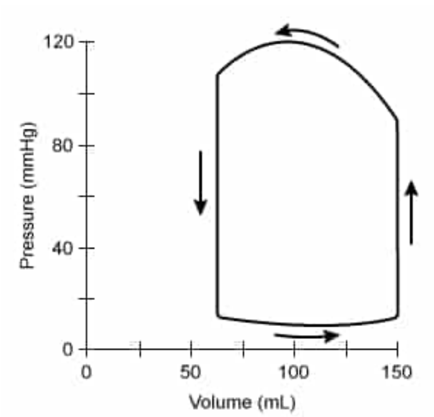
Equation 2

where *r* is the radius of the blood vessel and *L* is its length.

The pressure and volume changes that occur in the left ventricle during a single heartbeat are plotted in the graph shown.



Given this information, which graphic best illustrates the pressure and volume changes that occur in the right ventricle?

A

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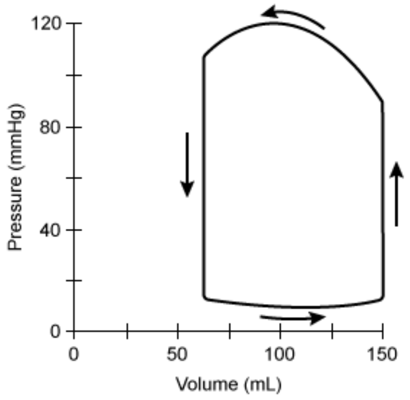
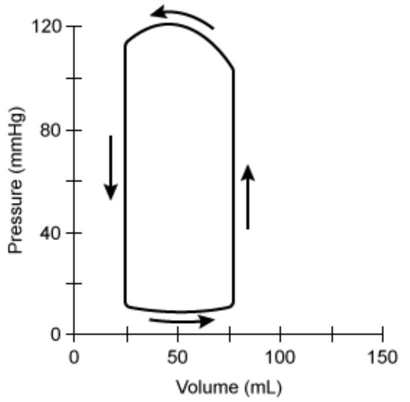
A.



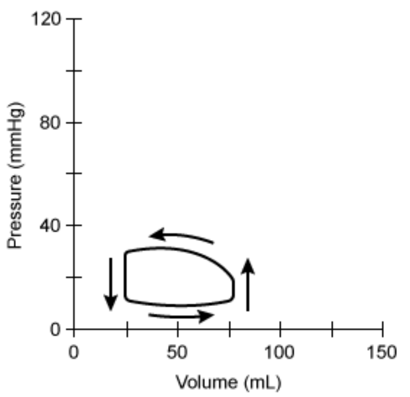
[8%]

B.



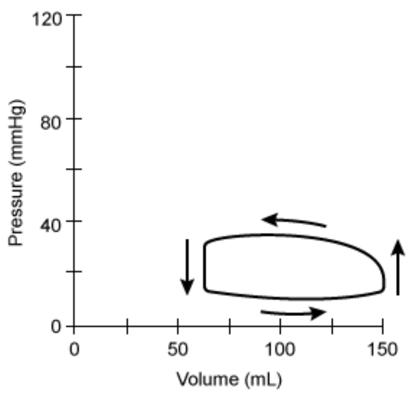
[6%]

C.



[8%]

D.



[75%]

Correct
75% answered correctly

Explanation:

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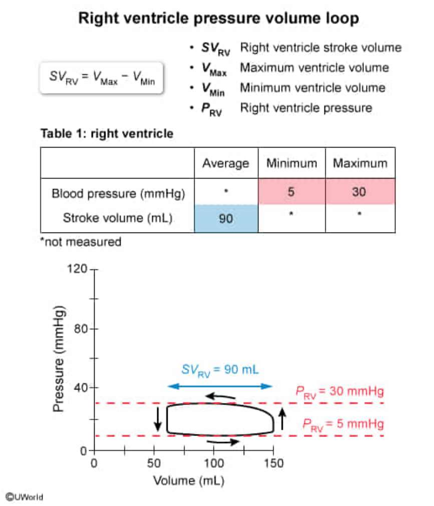

A plot of the pressure and volume within a chamber of the heart during the cardiac cycle is called a PV loop. The plot contains important information, such as the **stroke volume** *SV*, maximum pressure P_{Max}, and minimum pressure P_{Min} within the chamber of the heart. The *SV* is the difference between the maximum volume and the minimum volume in the PV loop:

$$\text{SV} = V_{\text{Max}} - V_{\text{Min}}$$

In this question, the PV loop of the left ventricle shows a stroke volume SV_{LV} of:

$$\text{SV}_{\text{LV}} = 150\text{ mL} - 60\text{ mL} = 90\text{ mL}$$

Table 1 in the passage lists the SV_{LV} as 90 mL in agreement with the calculation above. Similarly, both Table 1 and the PV loop show that P_{Max} is 120 mmHg and P_{Min} is 10 mmHg for the left ventricle.

Furthermore, Table 1 suggests that the PV loop of the right ventricle should have a stroke volume SV_{RV} of 90 mL, P_{Max} of 30 mmHg, and P_{Min} of 5 mmHg. The SV_{RV} for the PV loop in Choice D is:

$$\text{SV}_{\text{RV}} = 150\text{ mL} - 60\text{ mL} = 90\text{ mL}$$

In addition, P_{Max} and P_{Min} in Choice D are 30 mmHg and 5 mmHg, respectively. Therefore, Choice D illustrates the PV loop for the right ventricle.

(Choices A and B) The maximum pressure in these PV loops is 120 mmHg, but Table 1 shows the maximum pressure in the right ventricle as only 30 mmHg.

(Choice C) The stroke volume in this PV loop is 50 mL, but Table 1 shows the stroke volume of the right ventricle as 90 mL.

Educational objective:
A pressure volume loop shows the pressure and volume in the heart during the cardiac cycle. Pressure volume loops from the right and left ventricles show similar stroke volumes but lower pressure in the right ventricle.

Time spent:0 sec
Subject: Physics

QID:403696
System: Fluids

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Researchers measured several cardiovascular parameters from the heart and aorta in patients undergoing cardiac catheterization. The values in Table 1 were measured by inserting catheters through the vascular systems of the patients and utilizing ultrasound imaging, pressure sensors, and Doppler velocity sensors. A catheter with a Doppler velocity sensor operating at an ultrasound frequency of 10 MHz was inserted into the aorta to measure blood velocity at different phases of the cardiac cycle.

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	Stroke volume (mL)	90	*	*
Right ventricle	Blood pressure (mmHg)	*	5	30
	Stroke volume (mL)	90	*	*

*not measured

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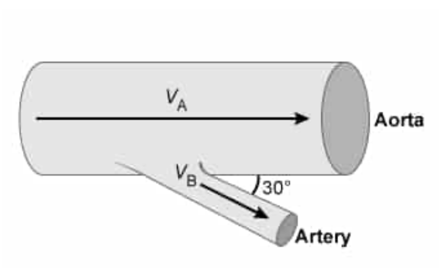
Equation 2

where *r* is the radius of the blood vessel and *L* is its length.

An artery branches off the aorta at a 30° angle. The average blood velocity in the artery V_B is one-tenth of the average blood velocity in the aorta V_A.



What is the component of the average blood velocity in the artery that is perpendicular to the flow in the aorta? (Note: sin 30° = 0.5; cos 30° = 0.86.)

- A. 1.0 cm/s [53%]
- B. 2.0 cm/s [20%]
- C. 10 cm/s [17%]
- D. 17 cm/s [7%]

Correct
53% answered correctly

Explanation:

Perpendicular component of blood velocity in artery

$$V_B = \frac{V_A}{10}$$
$$V_{B\perp} = V_B \sin\theta$$

- V_B Blood velocity in artery
- V_A Blood velocity in aorta
- $V_{B\perp}$ Perpendicular component of V_B
- θ Angle of artery

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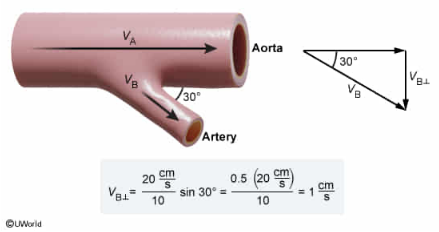

A vector is characterized by both its magnitude and direction. In two dimensions, a vector can be separated into two components that align with orthogonal axes, typically denoted the *x*- and *y*-axis. The overall (resultant) vector V forms an angle θ with the *x*-axis and represents the hypotenuse of a right triangle. The *x*-component forms the adjacent side of the right triangle, and the *y*-component forms the opposite side of the triangle.

Consequently, the trigonometric ratios of right triangles can be used to calculate the magnitude of the component vectors along the *x*-axis V_x and the *y*-axis V_y:

$$V_x = V\cos\theta$$

$$V_y = V\sin\theta$$

In this question, the vector of blood **velocity** V_B in the artery is separated into its components relative to the direction of blood velocity V_A in the aorta. Therefore, the components of V_B are aligned parallel $V_{B\parallel}$ and perpendicular $V_{B\perp}$ to V_A:

$$V_{B\parallel} = V_B\cos\theta$$

$$V_{B\perp} = V_B\sin\theta$$

The magnitude of V_B is one-tenth the magnitude of V_A, yielding:

$$V_{B\perp} = \frac{V_A}{10}\sin\theta$$

Table 1 in the passage shows that V_A is 20 cm/s, and the question states that θ is 30°. Hence:

$$V_{B\perp} = \frac{20\ \frac{cm}{s}}{10}\sin 30° = \frac{0.5\left(20\ \frac{cm}{s}\right)}{10} = 1\ \frac{cm}{s}$$

Therefore, $V_{B\perp}$ equals 1 cm/s.

(Choice B) 2 cm/s equals the blood velocity in the artery, but the question asks for the perpendicular component of the blood velocity.

(Choice C) 10 cm/s is obtained by using the blood velocity in the aorta instead of the blood velocity in the artery.

(Choice D) 17 cm/s is the product of the blood velocity in the aorta and the cosine of 30°.

Educational objective:
The components of the blood velocity vector can be calculated from the magnitude of the resultant velocity vector and the angle of the resultant velocity vector relative to the horizontal axis.

Time spent:0 sec
Subject:
Physics

QID:403697
System:
Fluids

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	Stroke volume (mL)	90	*	*
Right ventricle	Blood pressure (mmHg)	*	5	30
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Equation 2

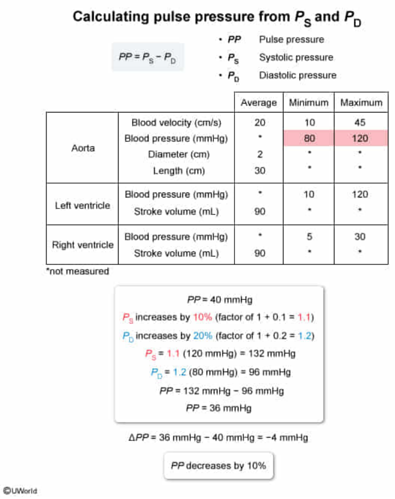
where *r* is the radius of the blood vessel and *L* is its length.

How will *PP* change if P_S increases by 10% and P_D increases by 20%?

- A. *PP* will increase by 20% [4%]
- B. *PP* will increase by 10% [16%]
- ✔ C. *PP* will decrease by 10% [73%]
- D. *PP* will decrease by 20% [5%]

Correct
72% answered correctly

Explanation:



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From Equation 1 in the passage, the definition of pulse pressure *PP* is the difference between the systolic pressure P_S and the diastolic pressure P_D:

$$\mathrm{PP} = P_\mathrm{S} - P_\mathrm{D}$$

The maximum blood pressure in the aorta equals P_S and the minimum blood pressure in the aorta equals P_D. Table 1 gives the maximum aortic blood pressure as 120 mmHg and the minimum aortic blood pressure as 80 mmHg. Hence, *PP* is calculated as:

$$P_\mathrm{S} = 120\ \mathrm{mmHg}$$

$$P_\mathrm{D} = 80\ \mathrm{mmHg}$$

$$\mathrm{PP} = 120\ \mathrm{mmHg} - 80\ \mathrm{mmHg} = 40\ \mathrm{mmHg}$$

In this question, P_S increases by 10% and P_D increases by 20%, yielding:

$$P_\mathrm{S} = 1.1(120\ \mathrm{mmHg}) = 132\ \mathrm{mmHg}$$

$$P_\mathrm{D} = 1.2(80\ \mathrm{mmHg}) = 96\ \mathrm{mmHg}$$

As a result, *PP* changes to:

$$\mathrm{PP} = 132\ \mathrm{mmHg} - 96\ \mathrm{mmHg} = 36\ \mathrm{mmHg}$$

Consequently, the change in pulse pressure Δ*PP* equals:

$$\Delta\ \mathrm{PP} = 36\ \mathrm{mmHg} - 40\ \mathrm{mmHg} = -4\ \mathrm{mmHg}$$

Furthermore, the percent change is the ratio of Δ*PP* and *PP*:

$$\frac{\Delta\ \mathrm{PP}}{\mathrm{PP}} = \frac{-4\ \mathrm{mmHg}}{40\ \mathrm{mmHg}} = -0.1 = -10\,\%$$

Therefore, *PP* decreased by 10%.

(Choice A) A 20% increase in pulse pressure is obtained if both the systolic and diastolic pressures increase by 20%, but the systolic pressure increased by only 10%.

(Choice B) A 10% increase in pulse pressure is obtained if both the systolic and diastolic pressures increase by 10%, but the diastolic pressure increased by 20%.

(Choice D) A 20% decrease in pulse pressure is obtained if only the diastolic pressure increased by 10%, but the systolic and diastolic pressures increased by 10% and 20%, respectively.

Educational objective:
Pulse pressure equals the difference between the systolic and diastolic blood pressures. The systolic pressure is the maximum blood pressure in the aorta, and the diastolic pressure is the minimum pressure in the aorta.

Time spent:0 sec
Subject:
Physics

QID:403698
System:
Fluids

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Equation 2

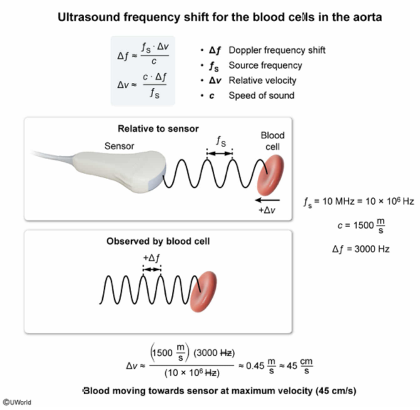
where *r* is the radius of the blood vessel and *L* is its length.

Blood cells in the aorta experience an ultrasound signal that is shifted +3000 Hz relative to the original signal transmitted from the Doppler velocity sensor. Which of the following best describes the aortic blood velocity in this situation? (Note: Assume speed of sound in the blood is *c* = 1500 m/s.)

- A. It is moving away from the Doppler velocity sensor at its minimum velocity [6%]
- B. It is moving toward the Doppler velocity sensor at its minimum velocity [19%]
- C. It is moving away from the Doppler velocity sensor at its maximum velocity [19%]
- ✔ D. It is moving toward the Doppler velocity sensor at its maximum velocity [54%]

Correct
53% answered correctly

Explanation:



Relative movement between a source and observer causes the ultrasound wave at the observer to differ in frequency *f* and wavelength *λ* compared to the original wave transmitted by the source. The change in frequency Δ*f* is called the

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Doppler shift and equals the difference between the observed frequency f_o and the source frequency f_s:

$$\Delta f = f_o - f_s$$

The value of Δf can be approximated from f_s, the **relative velocity** Δv, and the **speed of sound** c:

$$\Delta f \approx \frac{f_s \cdot \Delta v}{c}$$

This equation can be rearranged to solve for Δv, yielding:

$$\Delta v \approx \frac{\Delta f \cdot c}{f_s}$$

In this question, blood cells in the aorta experience a Δf of +3000 Hz and c equals 1500 m/s. The passage states that f_s of the sensor is 10 MHz (ie, 10×10^6 Hz). Substituting these values into the equation above yields:

$$\Delta v \approx \frac{(3000\ \cancel{\text{Hz}})\left(1500\ \frac{\text{m}}{\text{s}}\right)}{(10 \times 10^6\ \cancel{\text{Hz}})} \approx 0.45\ \frac{\text{m}}{\text{s}} \approx 45\ \frac{\text{cm}}{\text{s}}$$

With a positive Δf, Δv is also positive and the blood cells are moving toward the sensor. Furthermore, Δv is 45 cm/s, which is the maximum blood velocity in the aorta according to Table 1 in the passage. Therefore, the blood in the aorta is moving toward the sensor at its maximum velocity, corresponding to Choice D.

(Choices A and B) Table 1 in the passage states that the minimum blood velocity in the aorta is 10 cm/s, but the blood velocity in this question is 45 cm/s.

(Choice C) The Doppler shift would be negative if the blood cell were moving away from the sensor, but the Doppler shift is positive in this question.

Educational objective:
The Doppler shift can be approximated by the product of the frequency of the source and the ratio of the relative velocity and the speed of the wave in the medium, $\Delta f = (f_s \cdot \Delta v)/c$. When the source and observer move toward each other, the relative velocity and the Doppler shift are positive.

Time spent:0 sec
Subject:
Physics

QID:403699
System:
Fluids

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